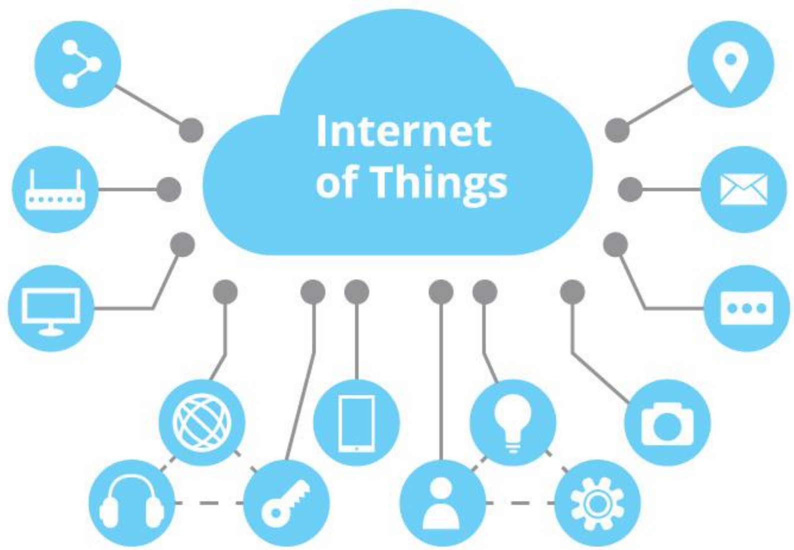
University of Economics Ho Chi Minh City
Institute of Intelligent & Interactive Technologies

Internet of Things (IoT)



Chapter 2: Hardware in IoT System

Pre: Nguyen Tan No

03/2025

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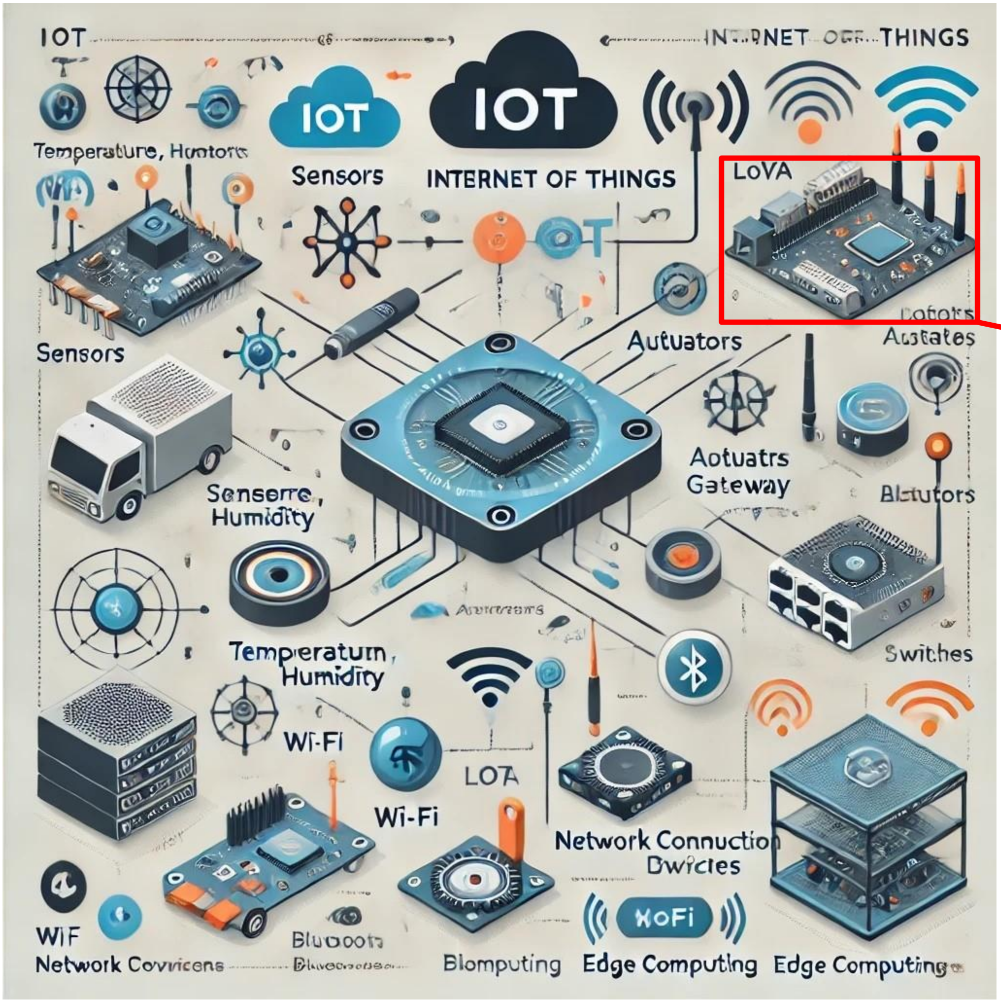
Overview of IoT System + Software + Basic Propgramming	Overview Embbeded System + Communication protocol	Sensors + Actutuators	Building an IoT System	Monitoring and Control System	Midterm Presentation			

Contents

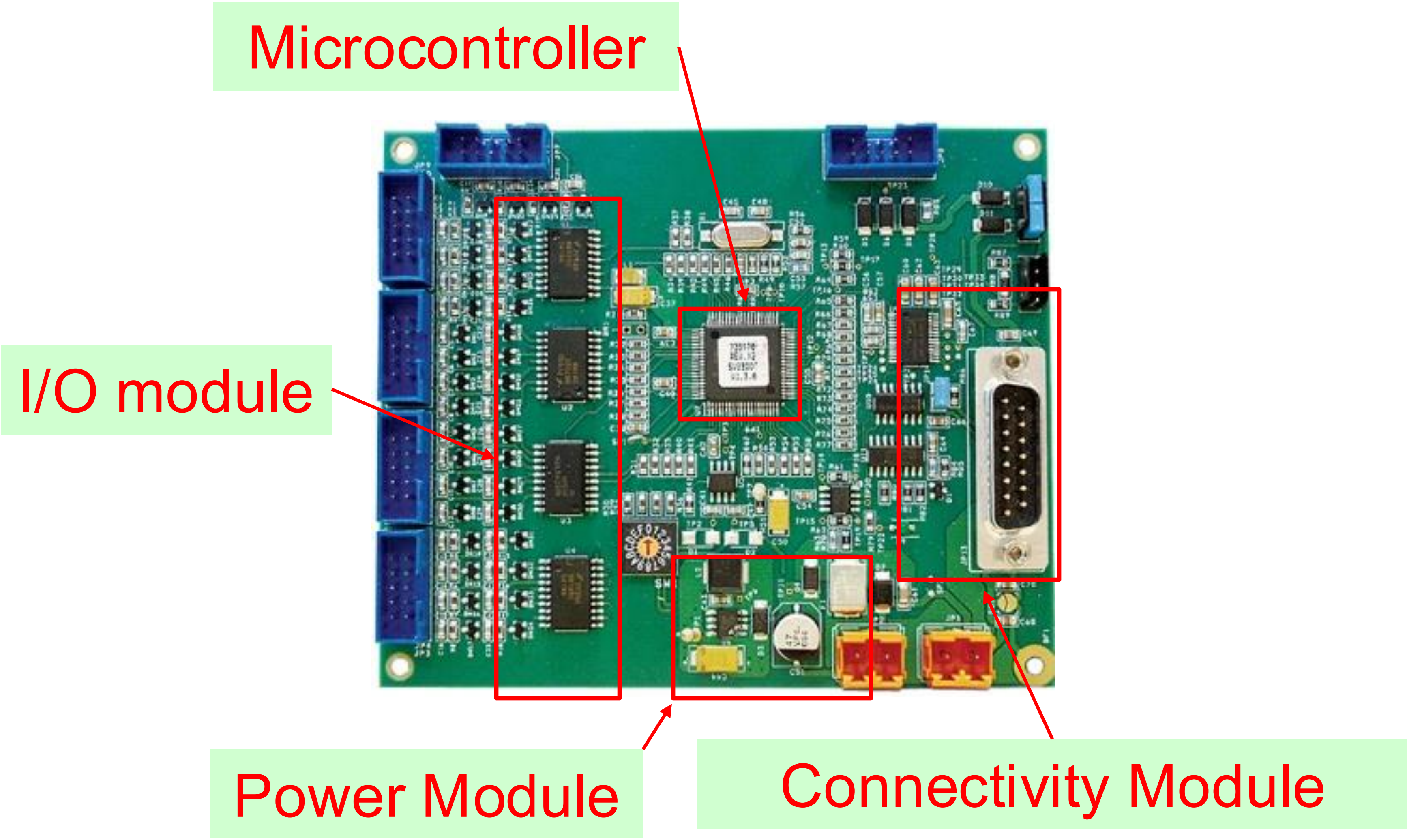
1. Embedded Systems



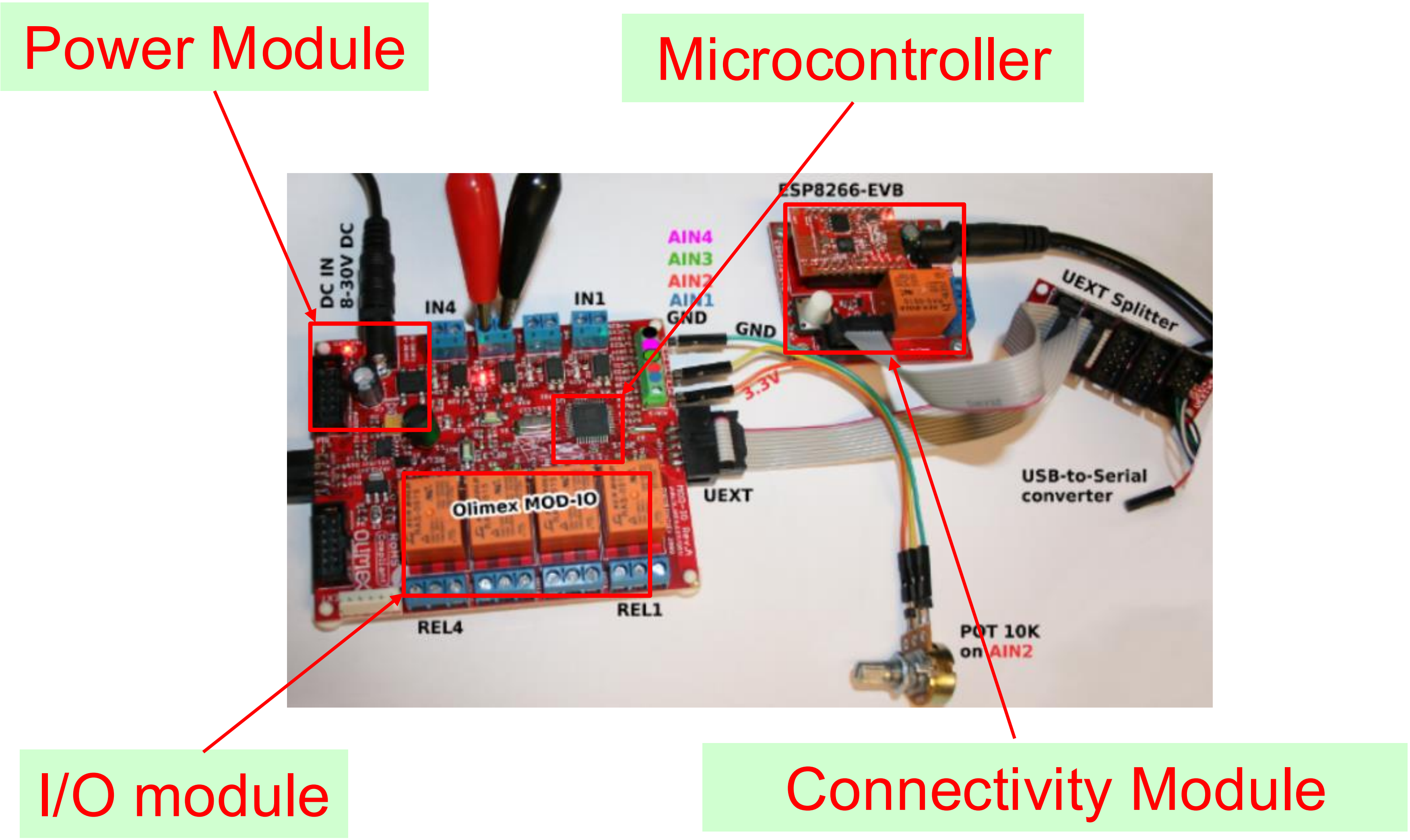
IoT System

IoT Module

1. Embedded Systems



1. Embedded Systems

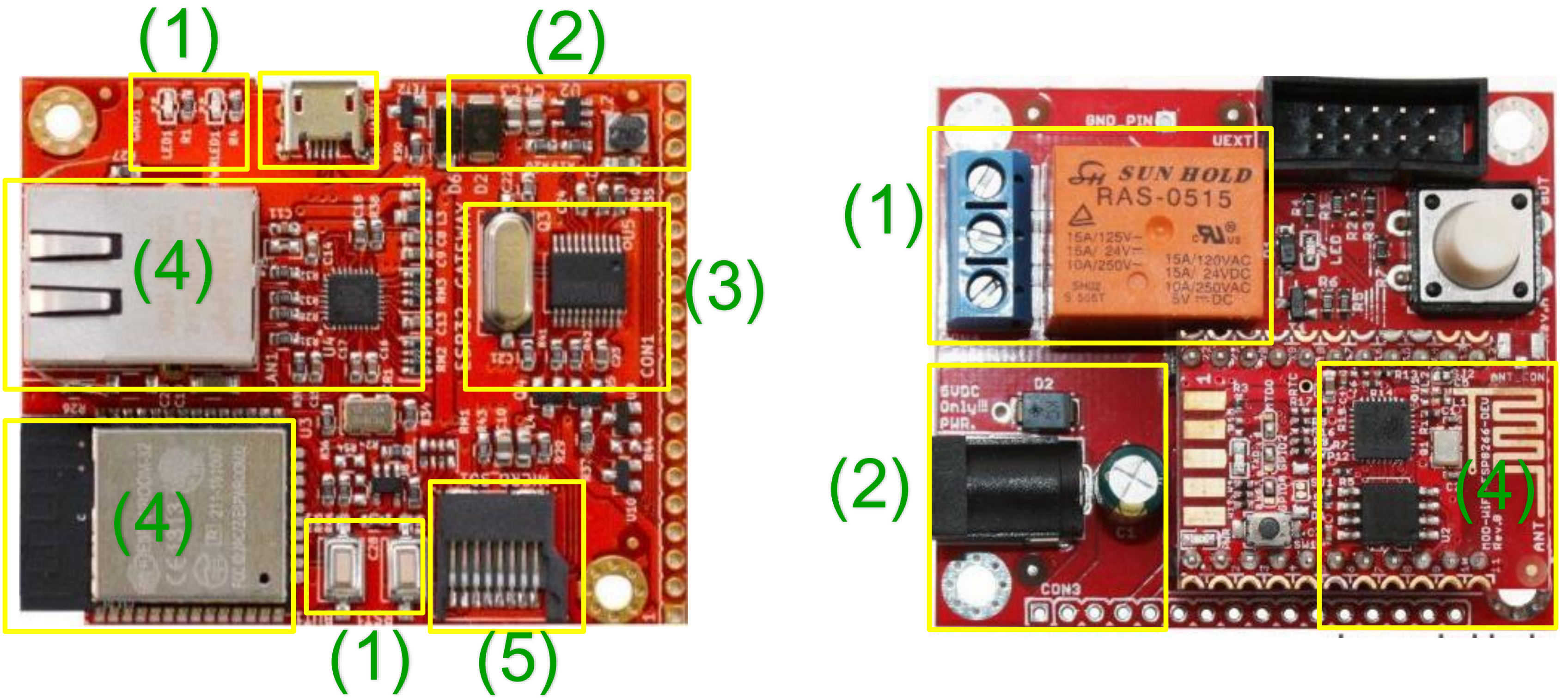


1. Embedded Systems

I/O module (1)

Power Module (2)

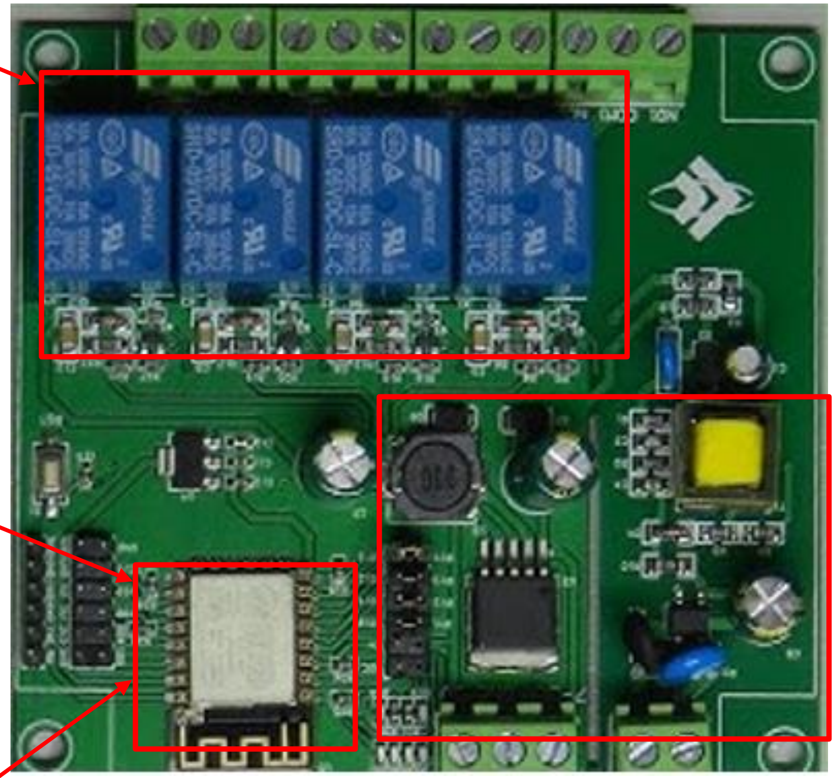
Microcontroller (3)



Connectivity Module (4)

Storage Module(5)

1. Embedded Systems

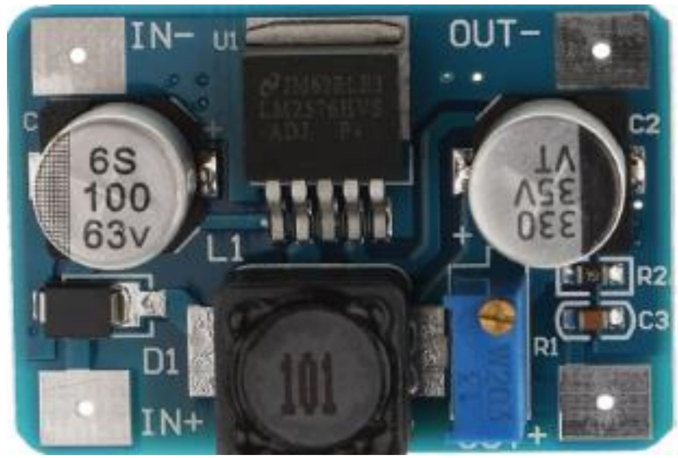


1. Embedded Systems

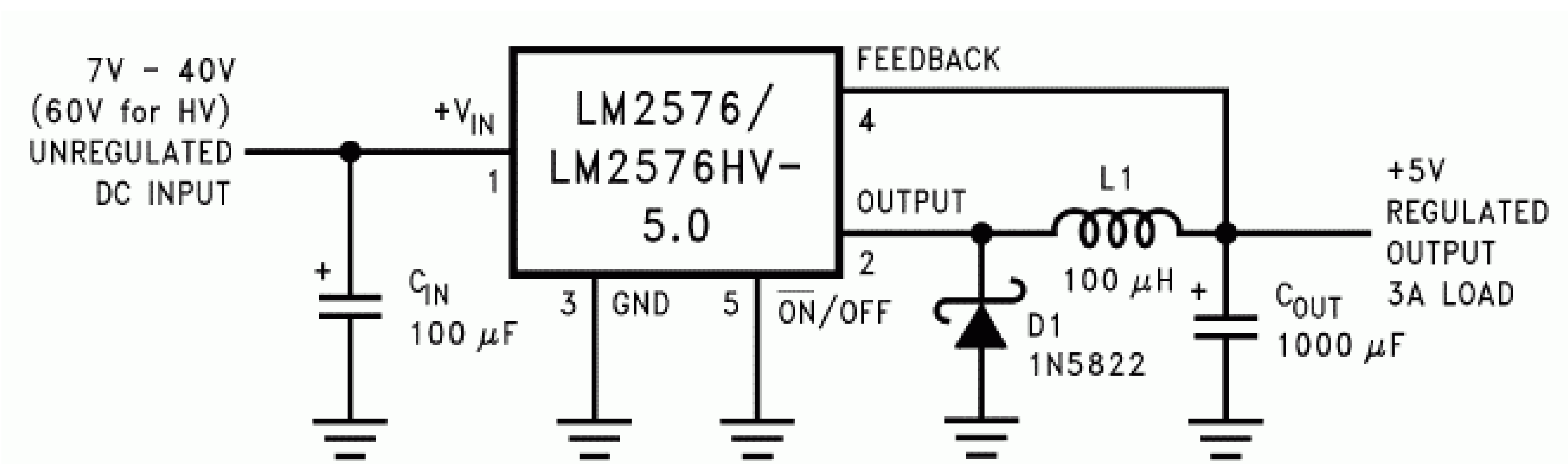
1. Power Supply.
2. Microcontroller/Microprocessor.
3. Sensors/Actuators.
4. Display Devices.
5. Storage Devices.
6. Connectivity Modules.
7. Gateway.

1. Embedded Systems: Power Supply

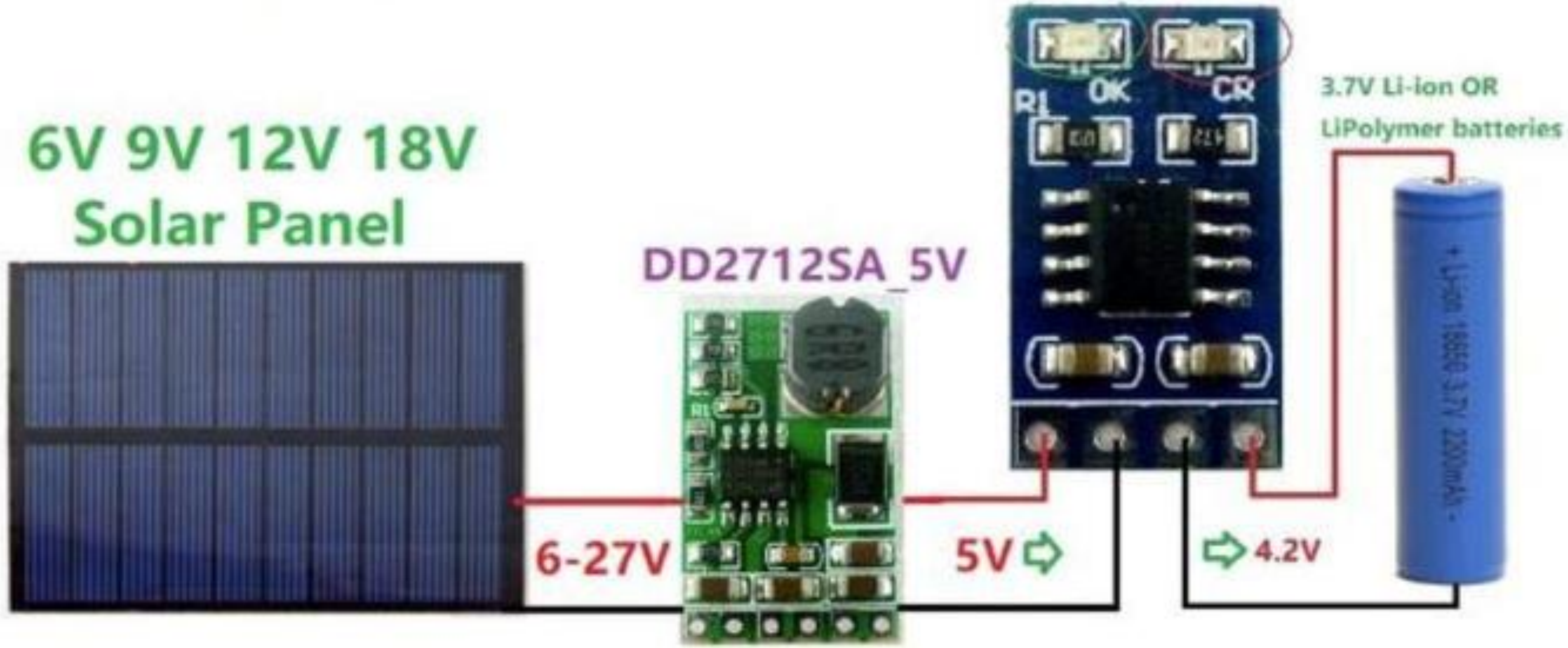
24Vdc-DC



5Vdc-3.3Vdc



1. Embedded Systems: Power Supply.

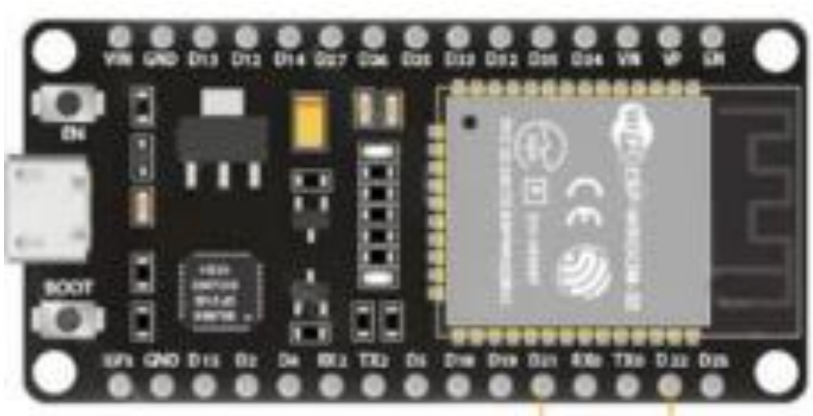


1. Embedded Systems: Microcontroller Microprocessor.

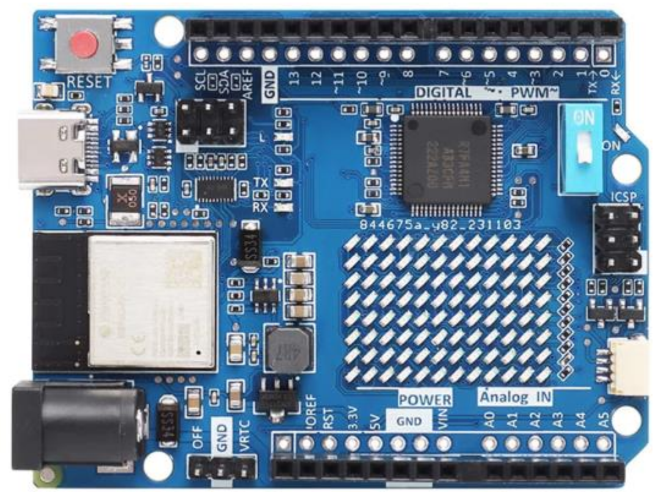
Esp32

STM32

ATMEGA328

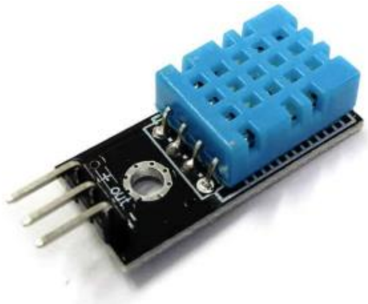


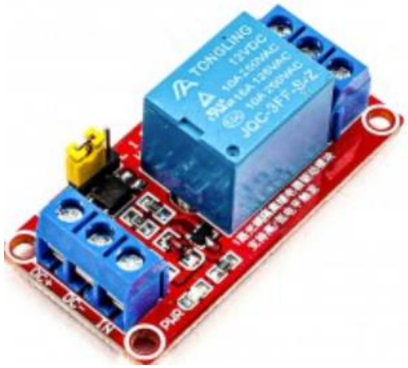
Esp32 Dev Kit

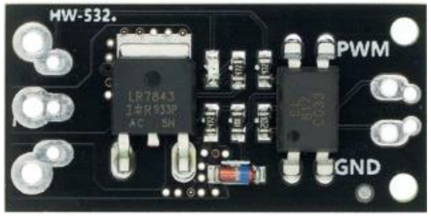


Esp32 Dev Kit

1. Embedded Systems: Sensors/Actuators.



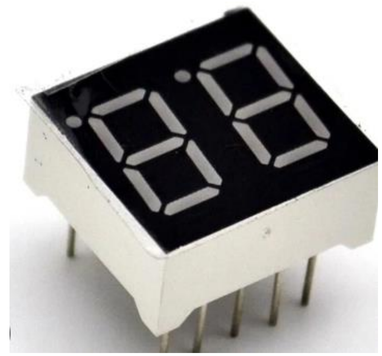


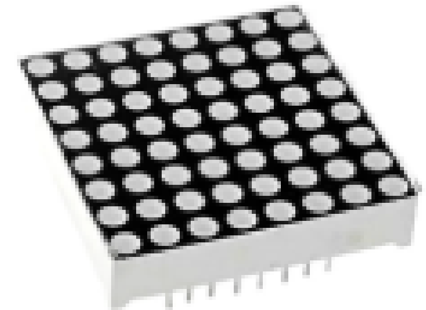


Sensors

Actuators

1. Embedded Systems: Display Devices.





Led

7 segment Led

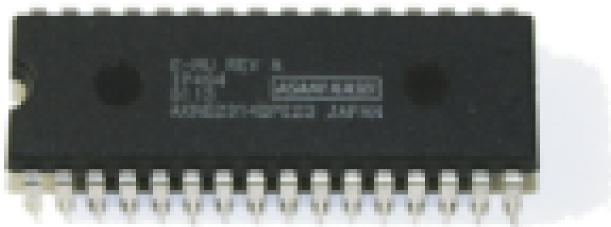
Matrix Led

LCD Display

Oled Display

RGB Oled

1. Embedded Systems: Storage Devices.

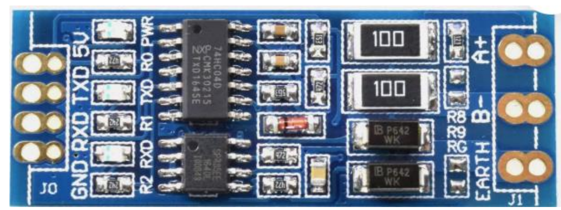


Memory IC

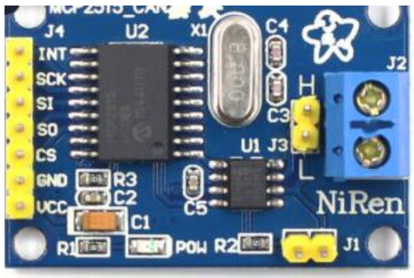
SD Card module

1. Embedded Systems: Connectivity Modules.

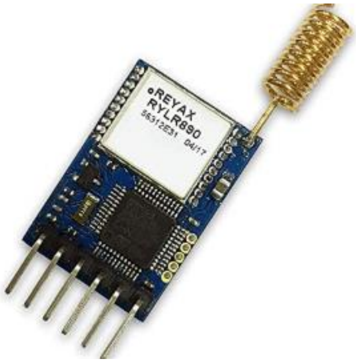

Ethernet Module



Rs485



CAN Bus

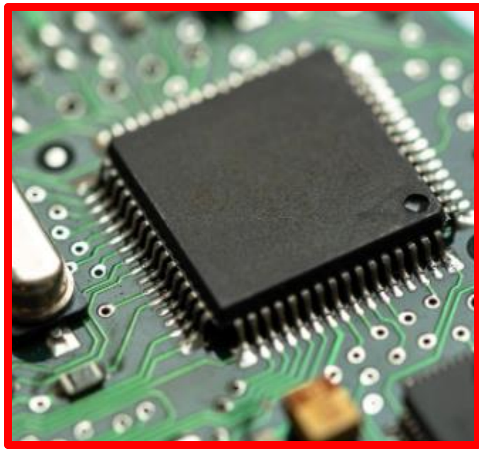

Lora


RF Module


Bluetooth


Remote

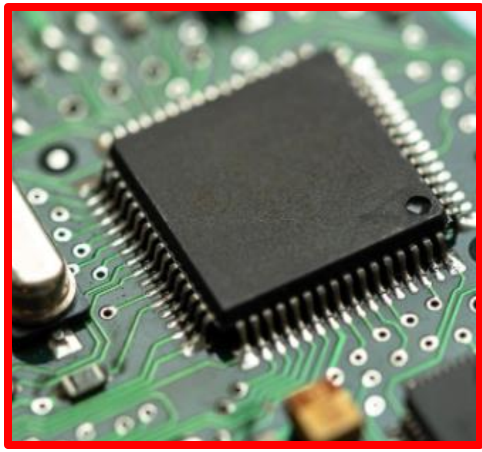
1. Embedded Systems:



Main Microcontroller

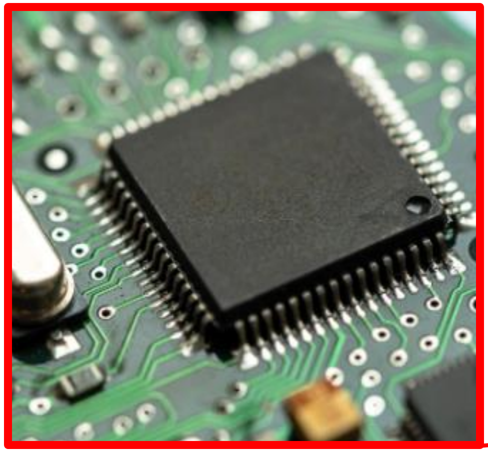
IoT Module

1. Embedded Systems:



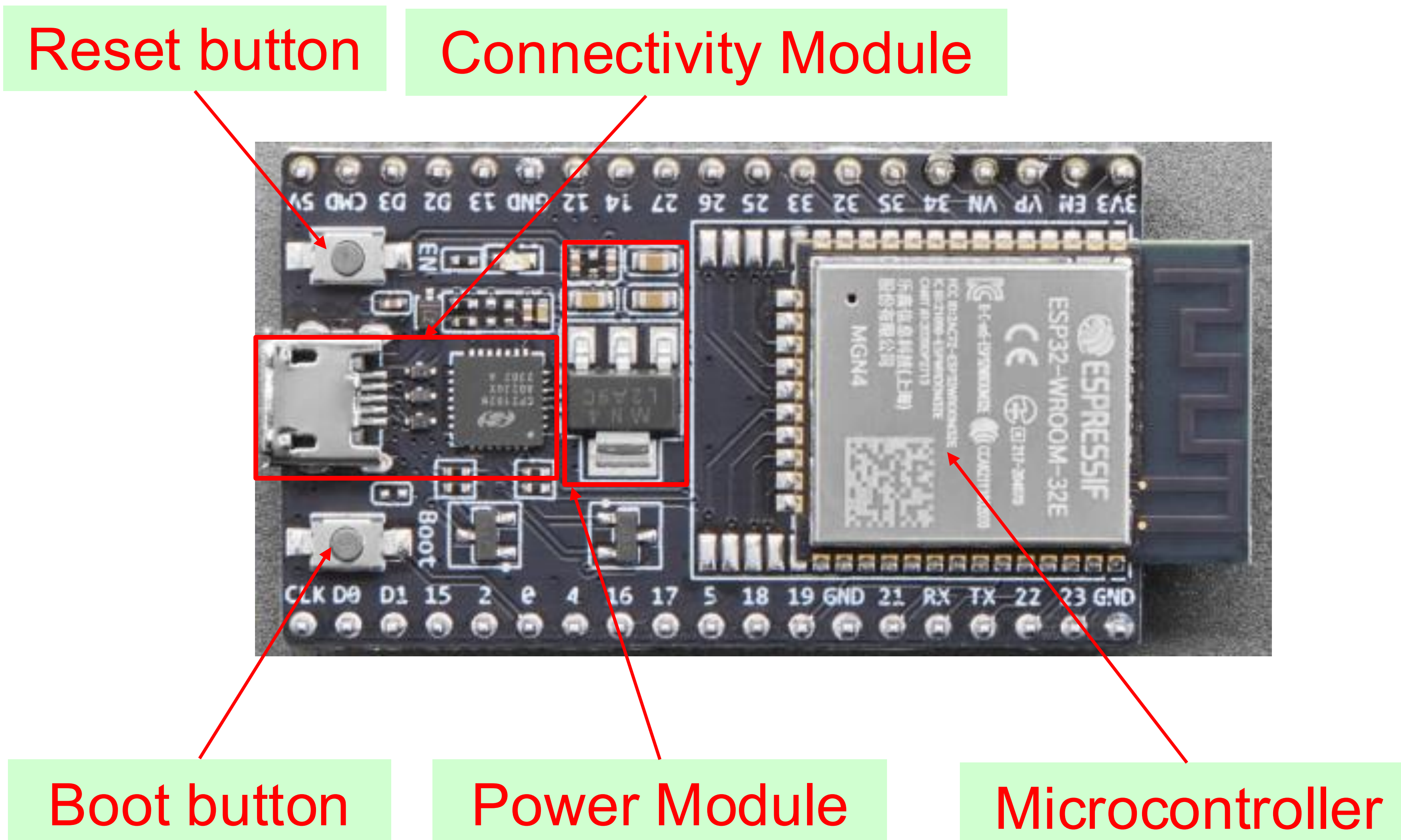
Client

1. Embedded Systems:



Host

1. Embedded Systems: Gateway (Cổng kết



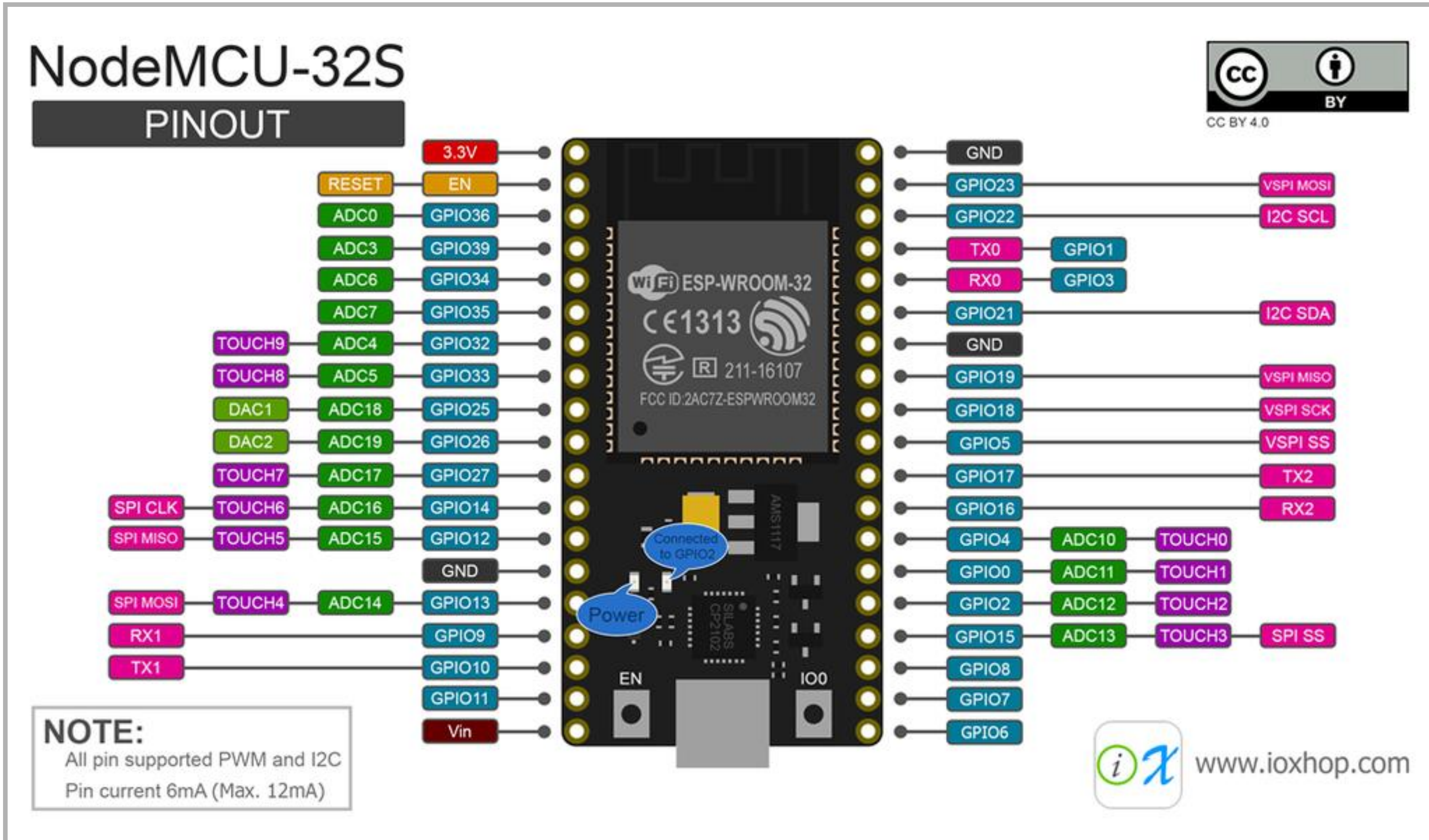
Esp32 Develop Kit

1. Embedded Systems

The ESP32 has the following peripherals:

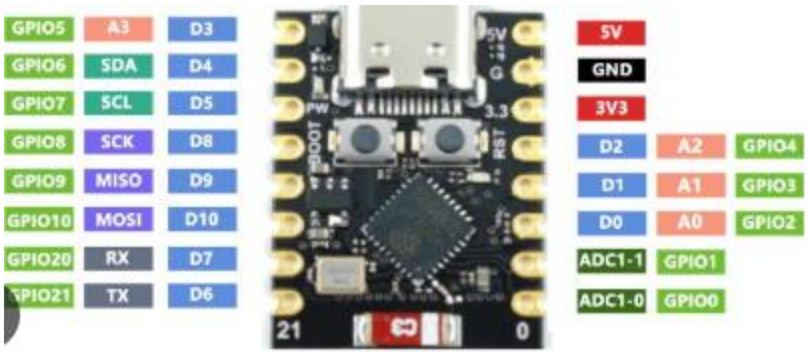
- 3 UART interfaces
- 2 I2C interfaces
- 3 SPI interfaces
- 16 PWM outputs
- 10 capacitive sensors
- 18 ADC channels
- 2 DAC outputs

1. Embedded Systems



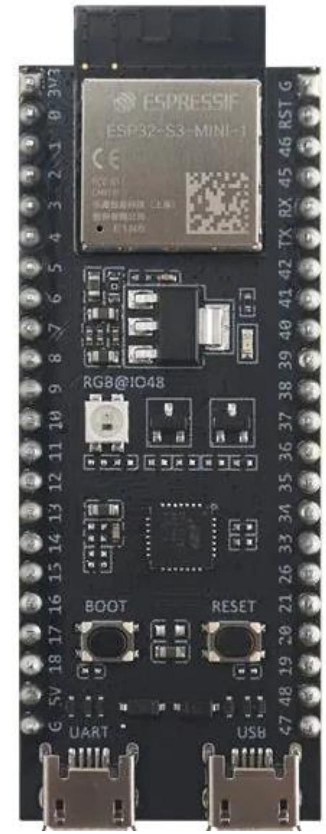
Esp32 Develop Kit

1. Embedded Systems

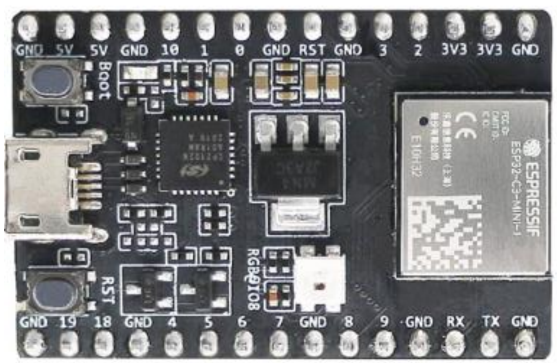


Esp32-C3

Esp32-S2

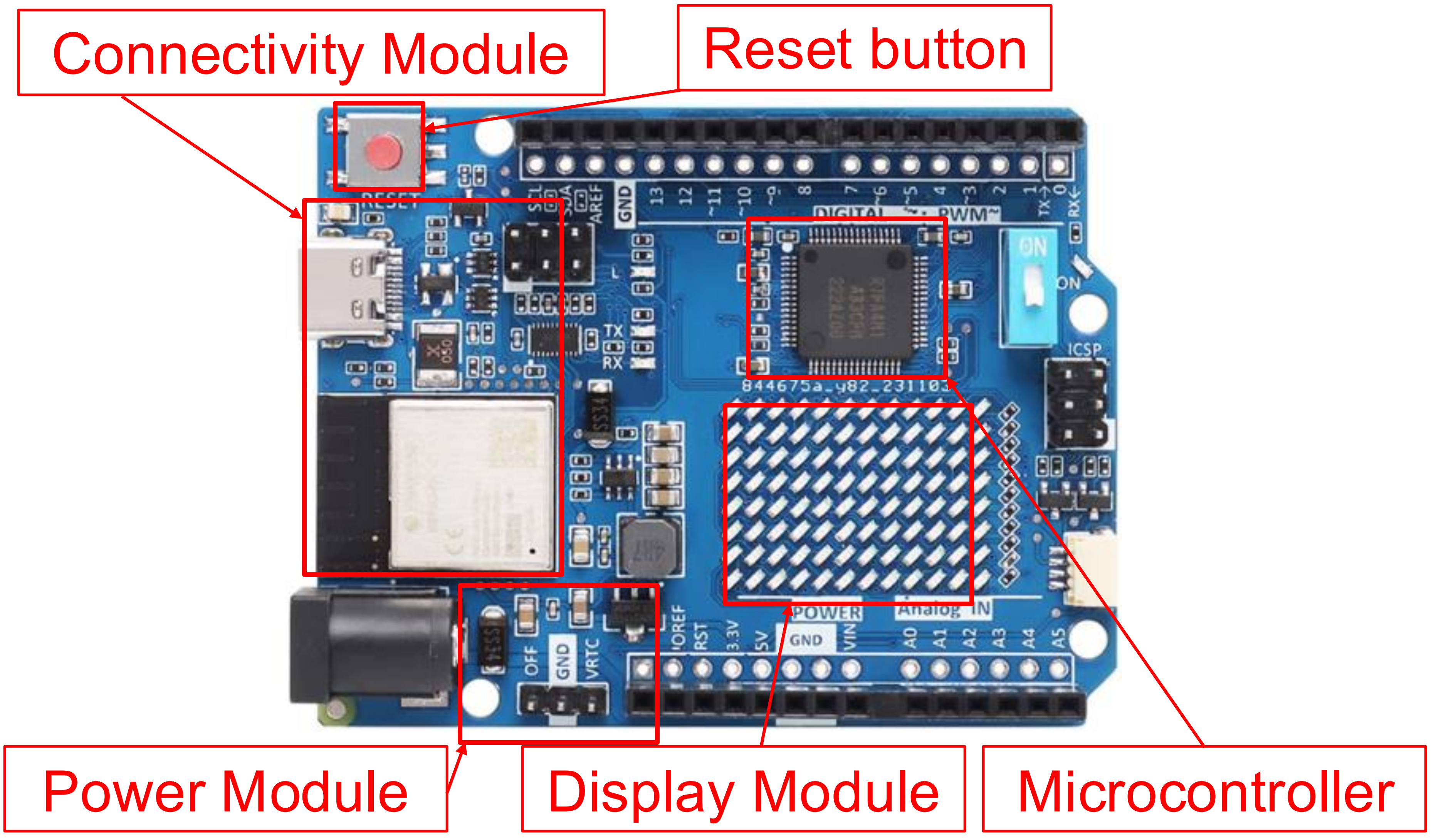


Esp32-S3



Esp32-C3

1. Embedded Systems

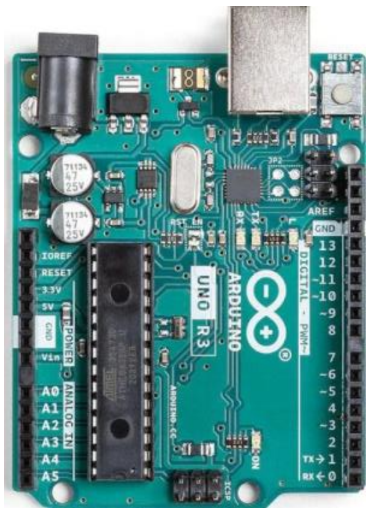


1. Embedded Systems

Arduino UNO R4 WIFI

Digital I/O Pins	14
Analog input pins	6
DAC	1
PWM pins	6
External interrupts	2,3
UART	Yes, 1x
I2C	Yes, 1x
SPI	Yes, 1x
CAN	Yes 1x CAN Bus
Circuit operating voltage	5 V (ESP32-S3 is 3.3 V)
Input voltage (VIN)	6-24 V
DC Current per I/O Pin	8 mA

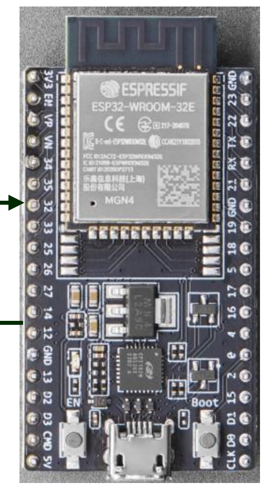
1. Embedded Systems



TX- RX

Serial Communication

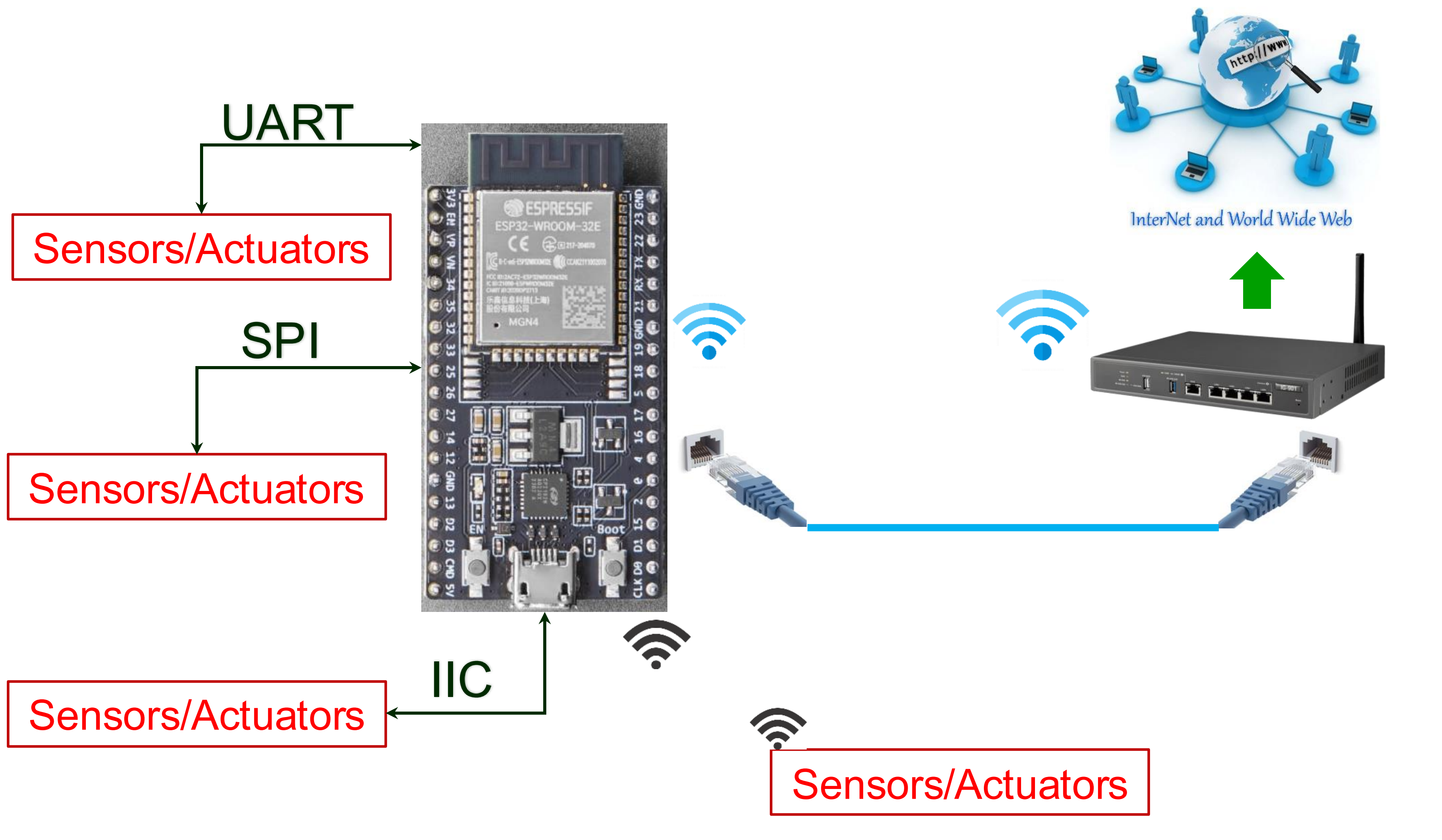
RX-TX



Arduino Uno

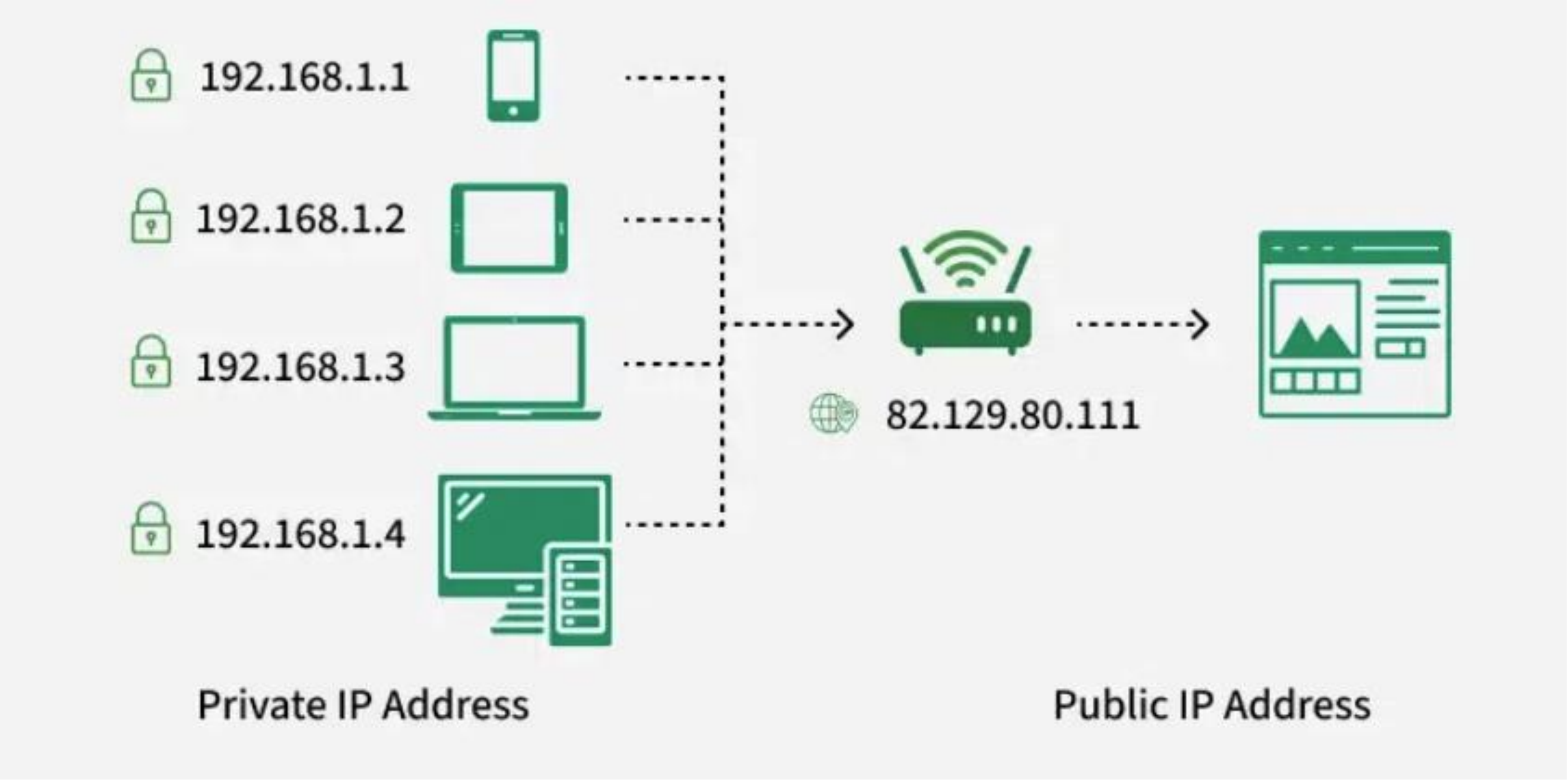
Esp32

2. Local Communication Protocols



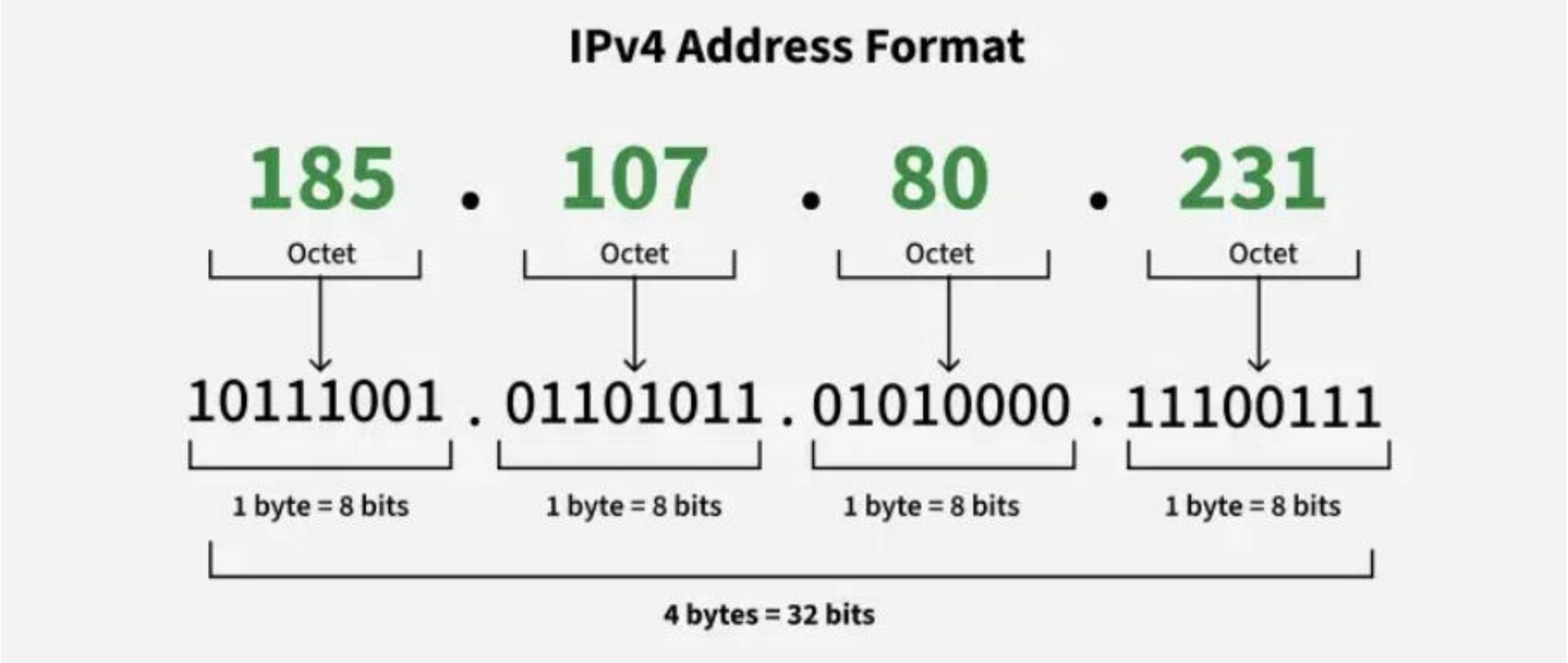
2. Local Communication Protocols

IP Adress



2. Local Communication Protocols

IP Adress



2. Local Communication Protocols

IP Adress

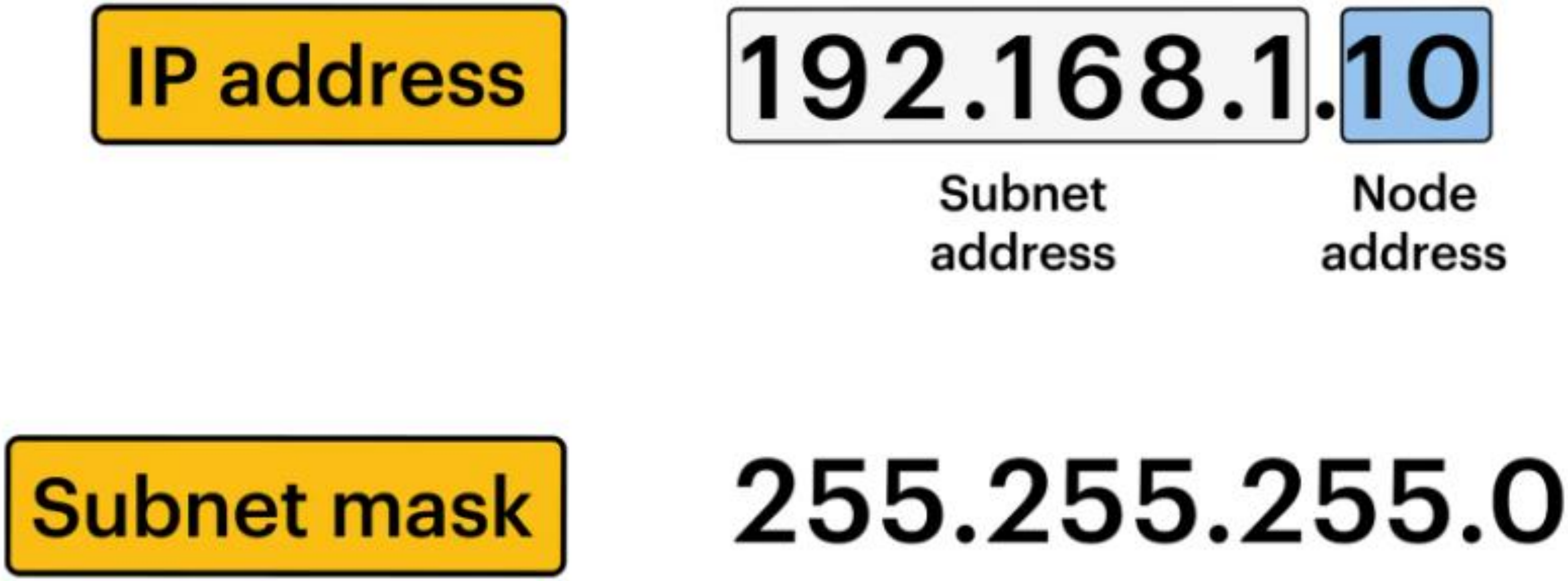
IPv6:

IPv6 addresses were created to deal with the shortage of IPv4 addresses. They use 128 bits instead of 32, offering a vastly greater number of possible addresses. These addresses are expressed as eight groups of four hexadecimal digits, each group representing 16 bits. The groups are separated by colons.

- **Example of IPv6 Address**: 2001:0db8:85a3:0000:0000:8a2e:0370:7334
    - Each group (like **2001**, **0db8**, **85a3**, etc.) represents a 16-bit block of the address.

2. Local Communication Protocols

Subnet mask



2. Local Communication Protocols

Subnet mask

Suppose you have the IP address **10.0.0.25** with a subnet mask of **255.255.255.240**.

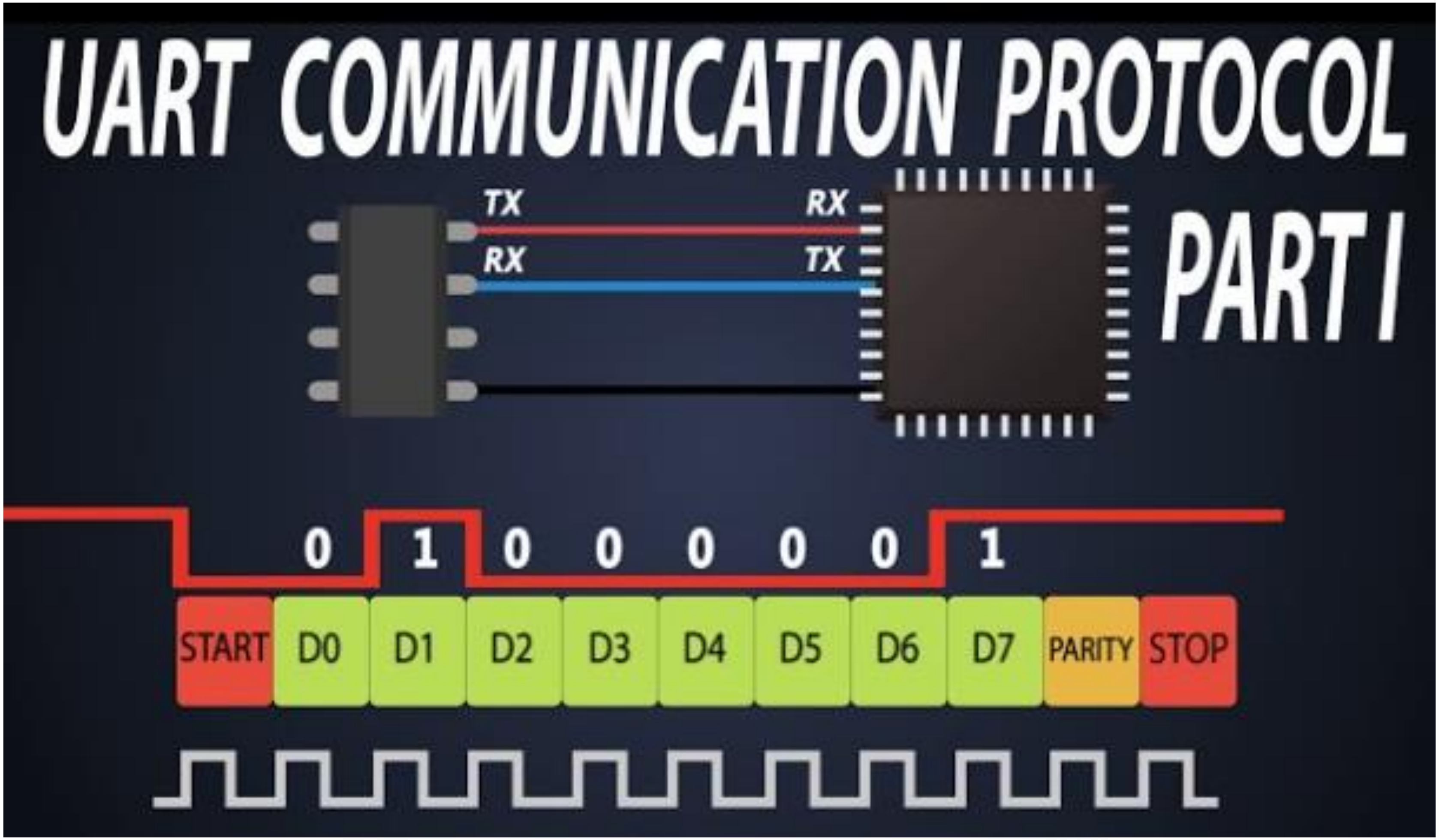
- **255.255.255.240** is equivalent to **/28** (28 bits for the network, 4 bits for the host).
- The total number of IP addresses in this subnet:

$$2^{(32-28)} = 2^4 = 16 \text{ addresses}$$

- Breakdown:
  - **Network Address: 10.0.0.16**
  - **Broadcast Address: 10.0.0.31**
  - **Usable IP range for hosts: 10.0.0.17 → 10.0.0.30**
  - **Total usable hosts: 14** (since 2 addresses are reserved for network and broadcast).

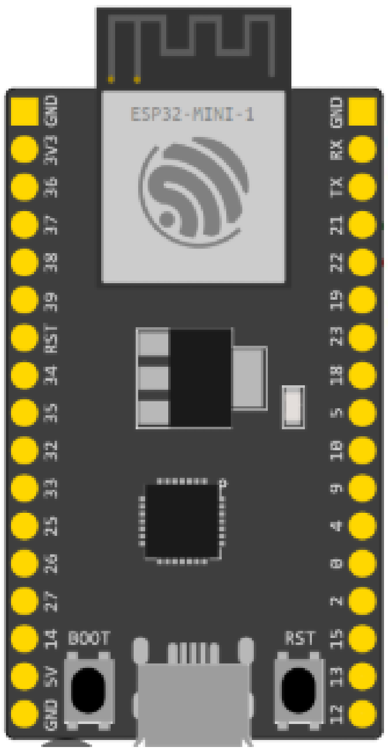
2. Local Communication Protocols

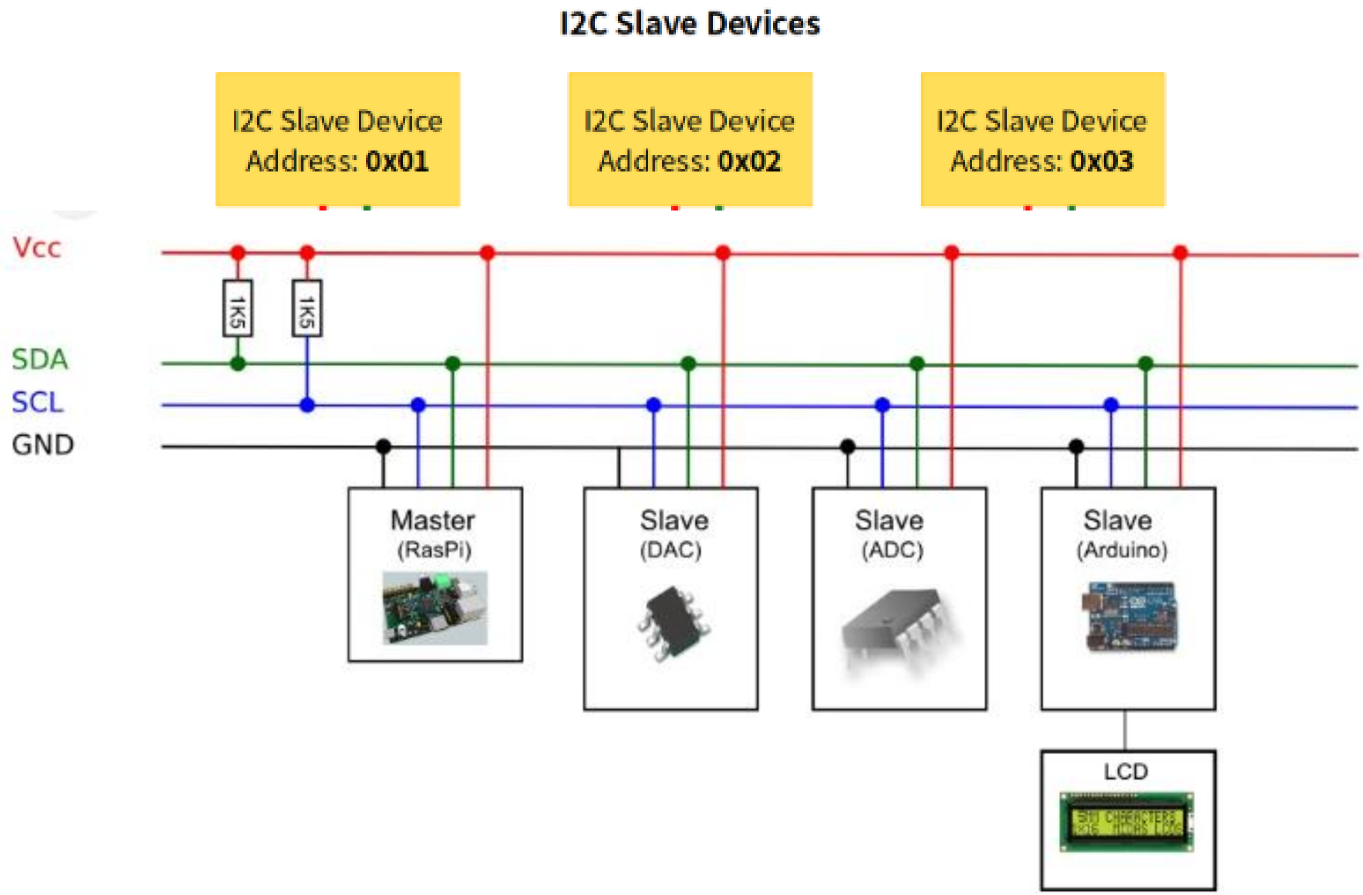
UART



2. Local Communication Protocols

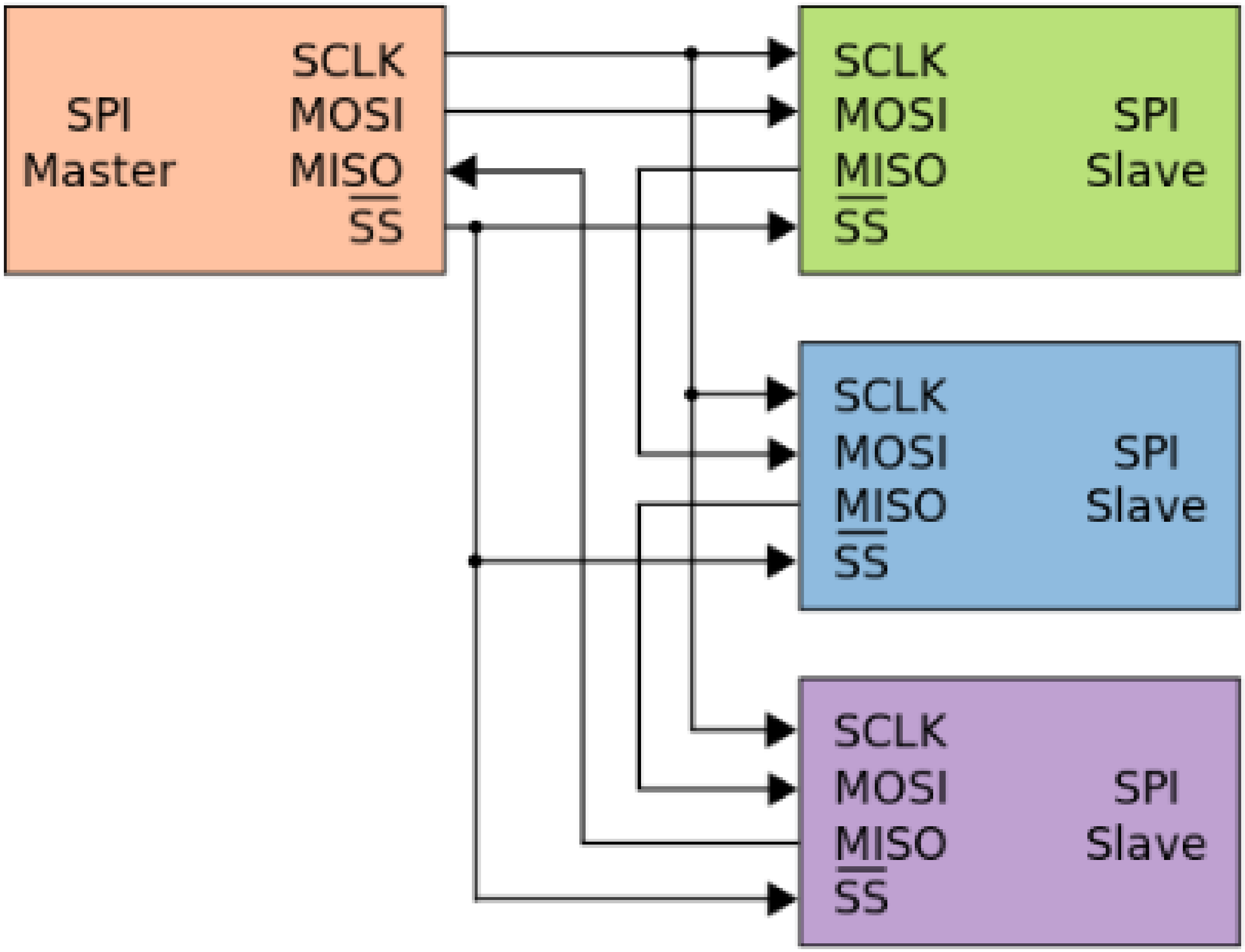
IIC





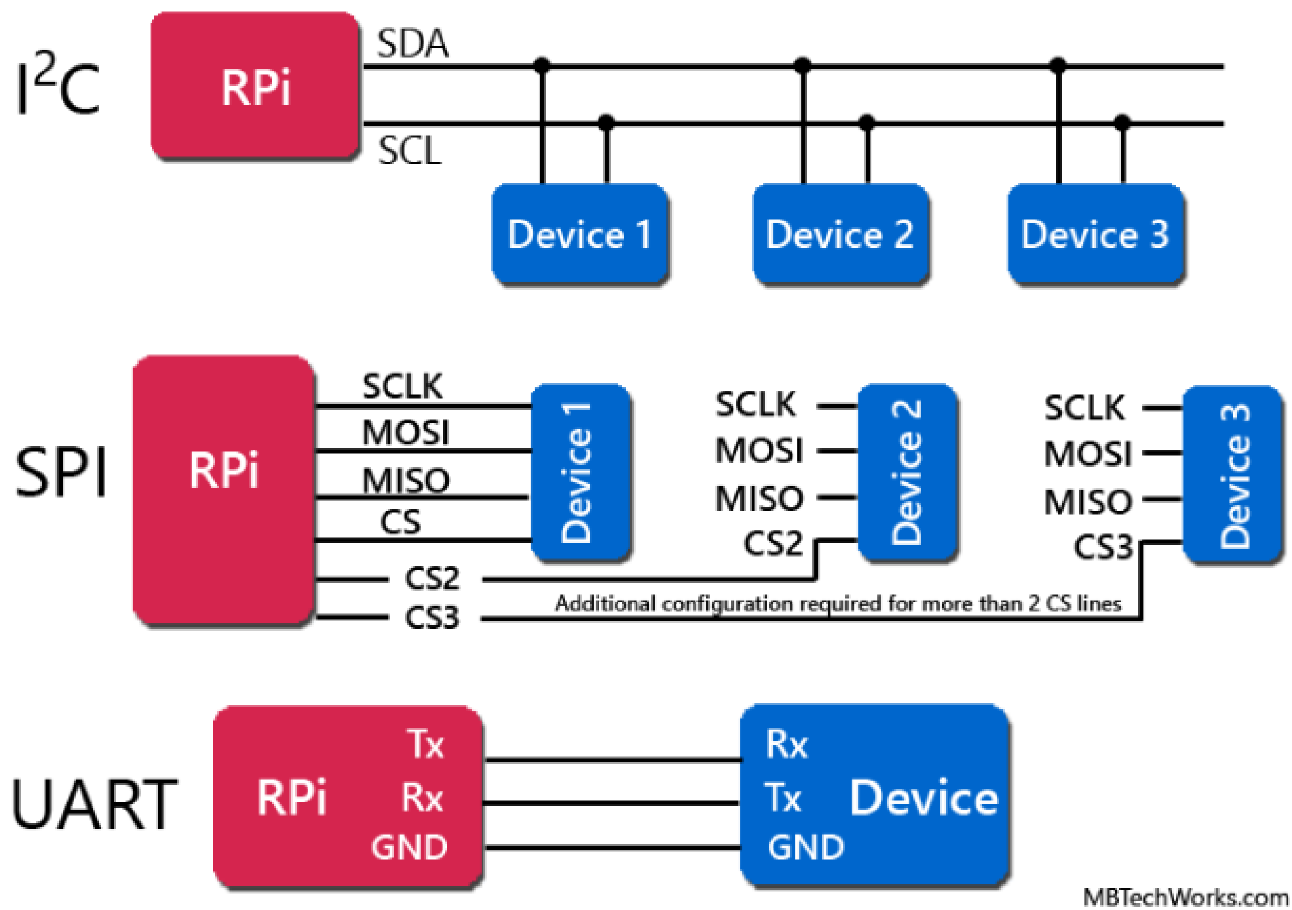
2. Local Communication Protocols

SPI



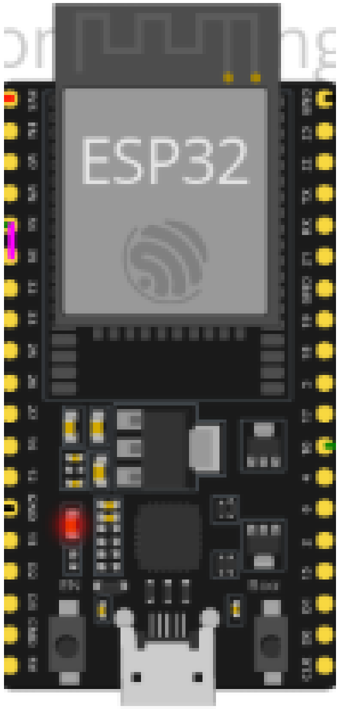
2. Local Communication Protocols

Compare

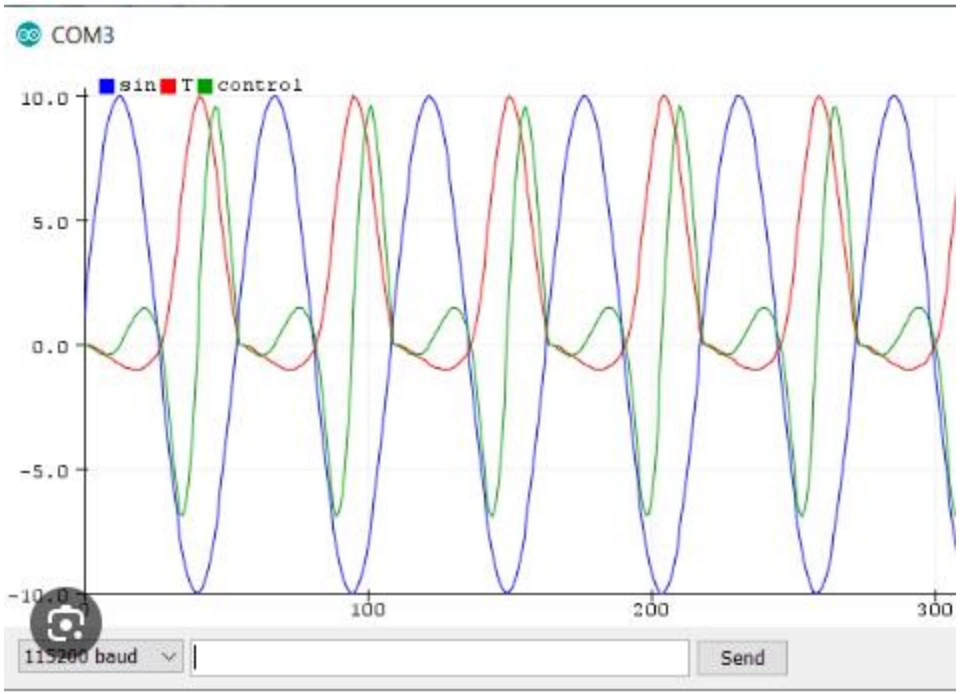


2. Local Communication Protocols

Practice 1



UART

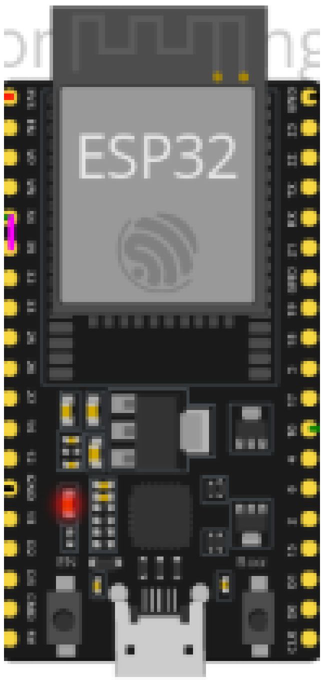


Esp32

Serial Plotter

2. Local Communication Protocols

Practice 2:



Wireless

Esp32-Host

Client

Thank for attention